The Effect of Truncation on the DFT Approximation to the FT

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Key Concepts

1) The effect of truncation is often the single dominant factor determining the quality of the DFT approximation to the Fourier transform.

2) Truncation of the signal to finite duration corresponds to multiplying the signal by a length-N rectangular window.

    a) Windowing corresponds to convolving the sampled signal discrete-time Fourier transform (DTFT) with the DTFT of the window.

b) The DTFT of the window has a main lobe of width $\frac{4\pi}{N}$ rads surrounded by side lobes whose peak amplitude are at -13 dB relative to the main lobe.

c) The main lobe blurs the detail in the underlying spectrum and limits the resolution of roughly equal amplitude components. It is difficult to resolve features that are significantly closer than the main lobe width $\frac{4\pi}{N}$ rads.

d) The side lobes limit the dynamic range. The side lobes due to a strong component cast a shadow throughout the entire spectrum. Weak amplitude spectral components are obscured by the side lobes of large amplitude spectral components.

Video Screenshots

Effect of Truncation on DTFT

– key step in using DFT to approximate the FT

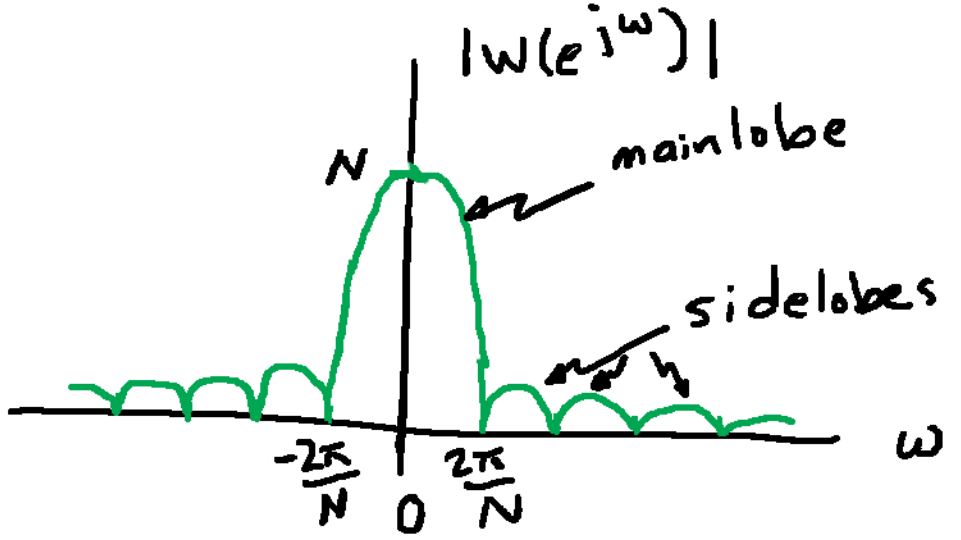
$z[n] = x[n]\,w[n]$ where $w[n] = \begin{cases} 1, & 0 \le n \le N-1 \\ 0, & \text{otherwise} \end{cases}$

$\updownarrow$ DTFT

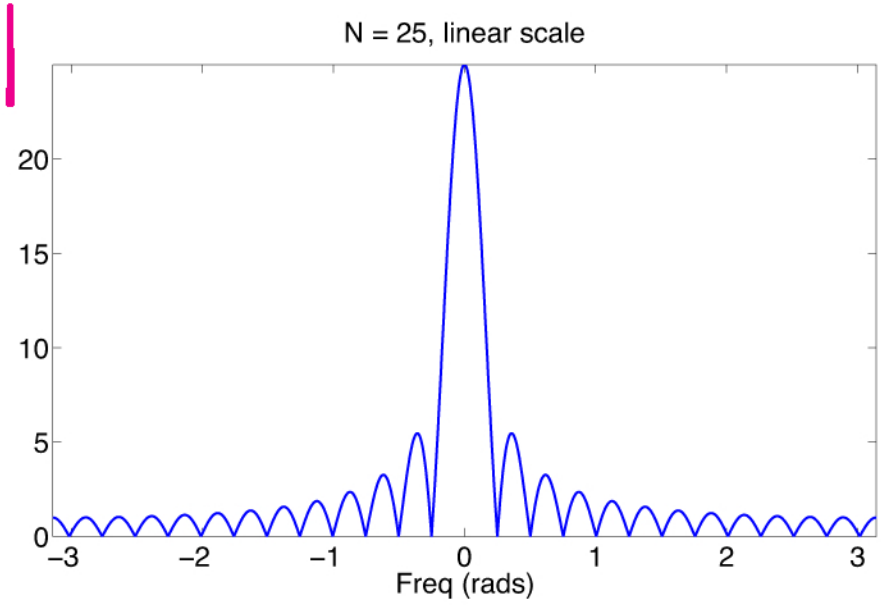
$$Z(e^{j\omega}) = \frac{1}{2\pi} X(e^{j\omega}) * W(e^{j\omega}) \quad ; \quad W(e^{j\omega}) = e^{-j\frac{(N-1)\omega}{2}} \frac{\sin(\frac{N\omega}{2})}{\sin(\omega/2)}$$

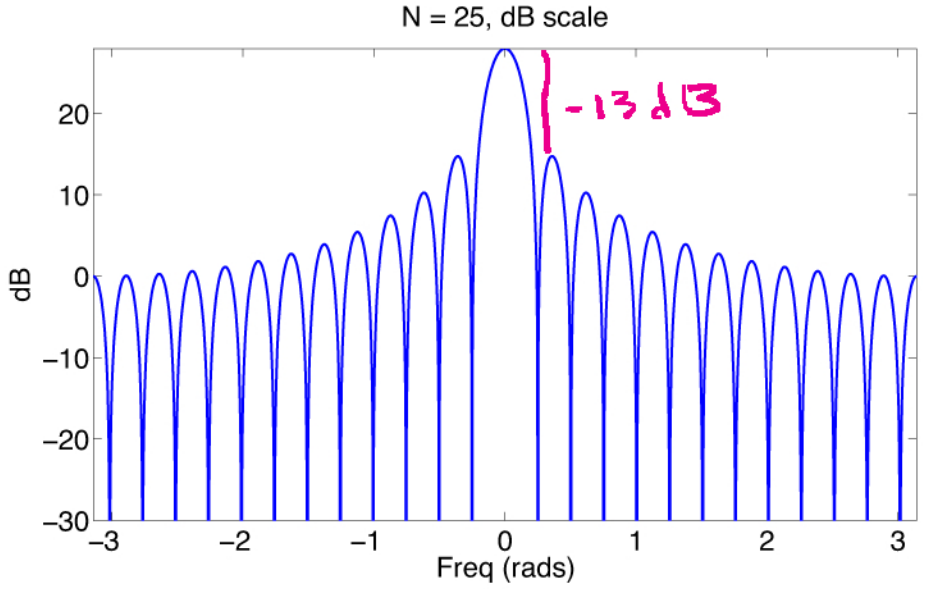
1) Mainlobe blurs detail in $X(e^{j\omega})$
Resolution ~ mainlobe width $= \frac{4\pi}{N}$

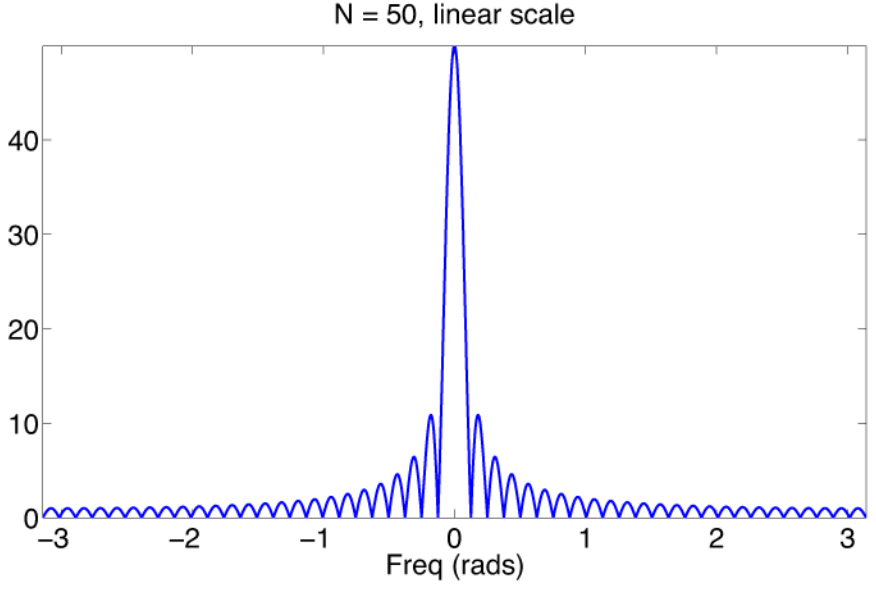
2) Sidelobes limit dynamic range.
Weak components obscured by sidelobes of large components

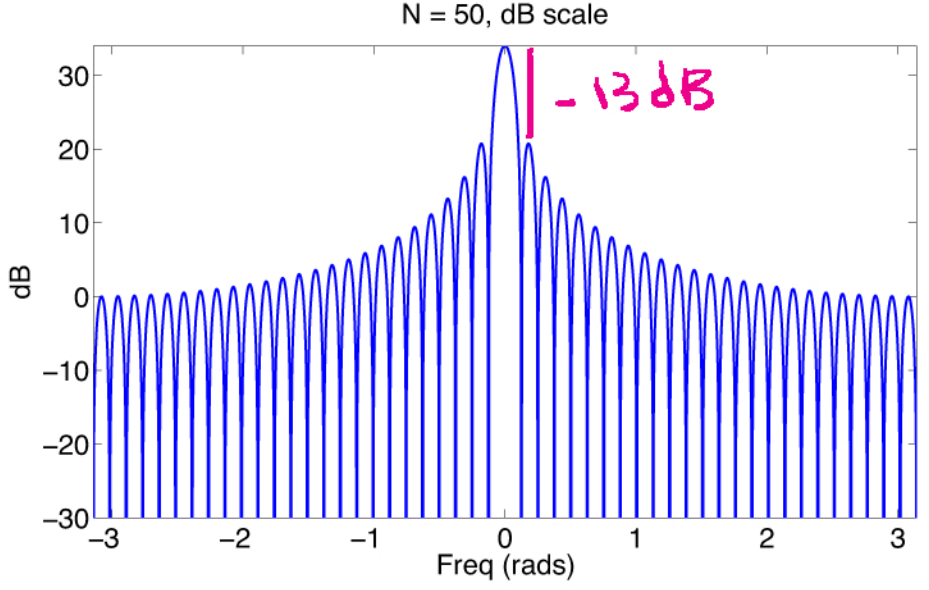


$|W(e^{j\omega})|$


N = 25, linear scale
Freq (rads)

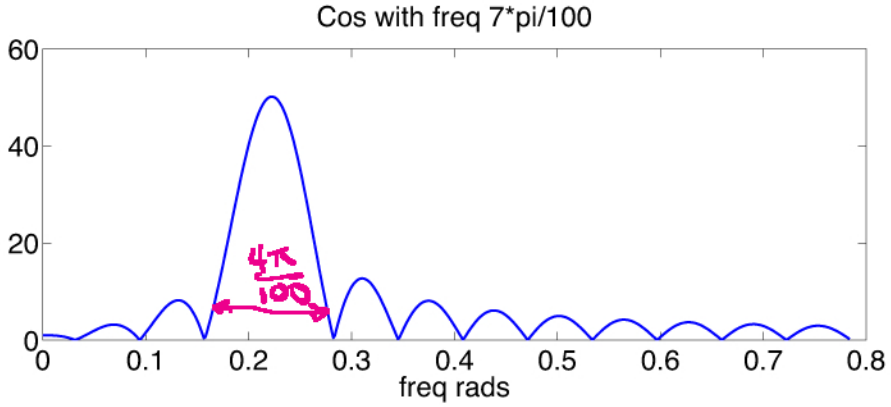

N = 25, dB scale
-13 dB
dB
Freq (rads)

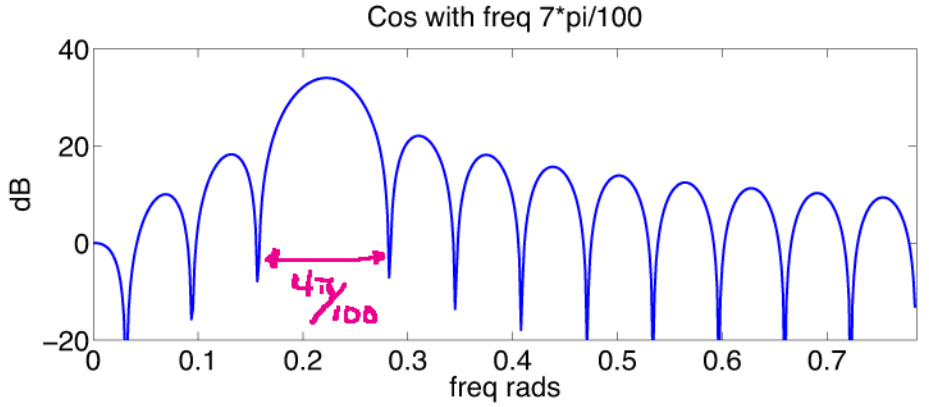

N = 50, linear scale
Freq (rads)

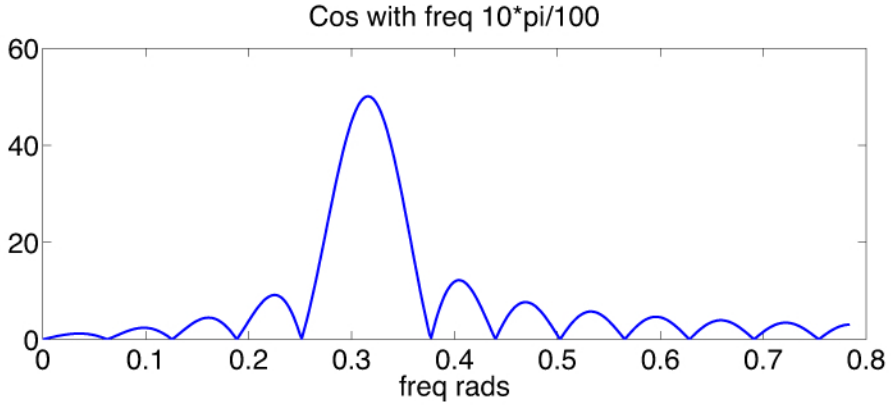

N = 50, dB scale
-13 dB
dB
Freq (rads)

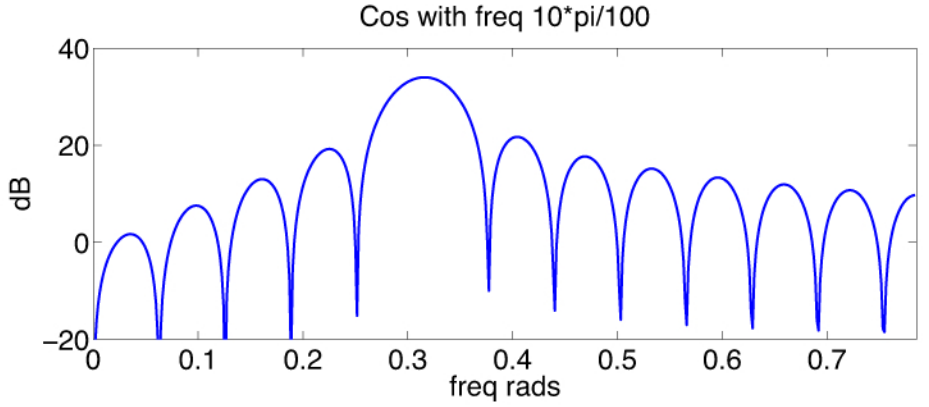
$|Z(e^{j\omega})|$

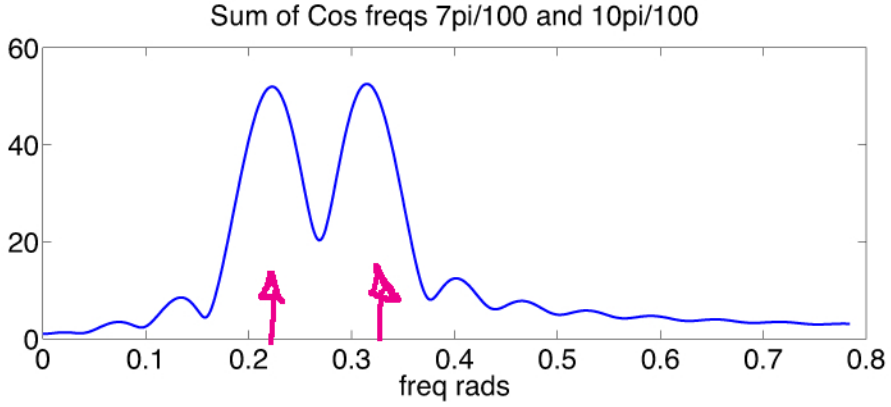
$N = 100$

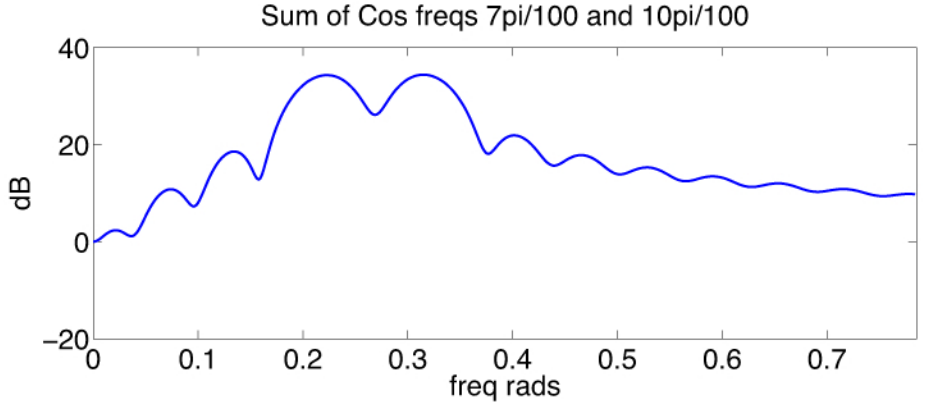








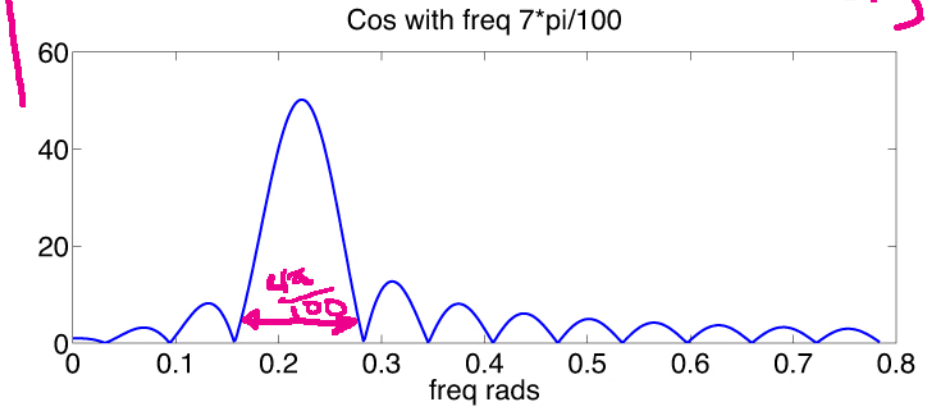


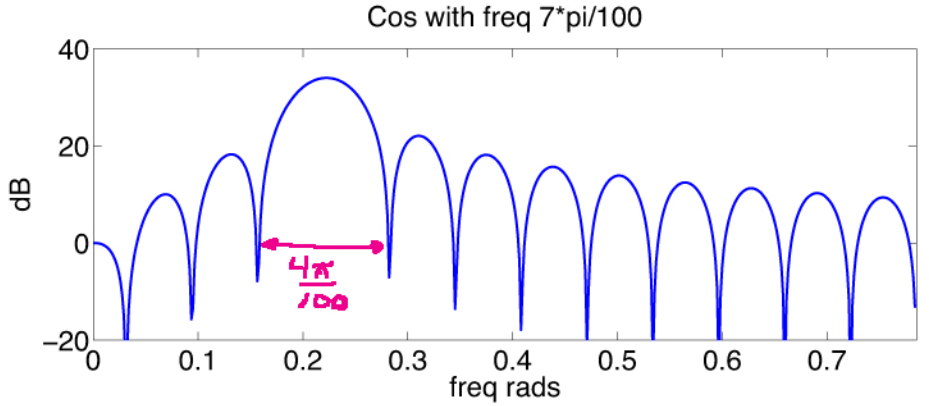


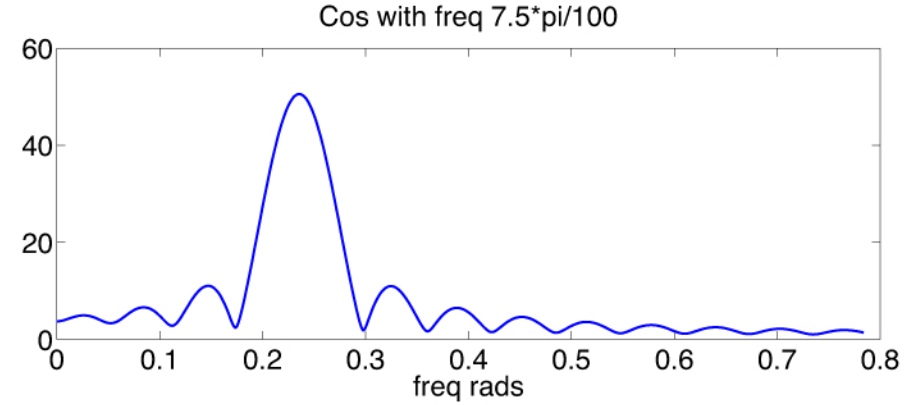
Closely Spaced Sinusoids

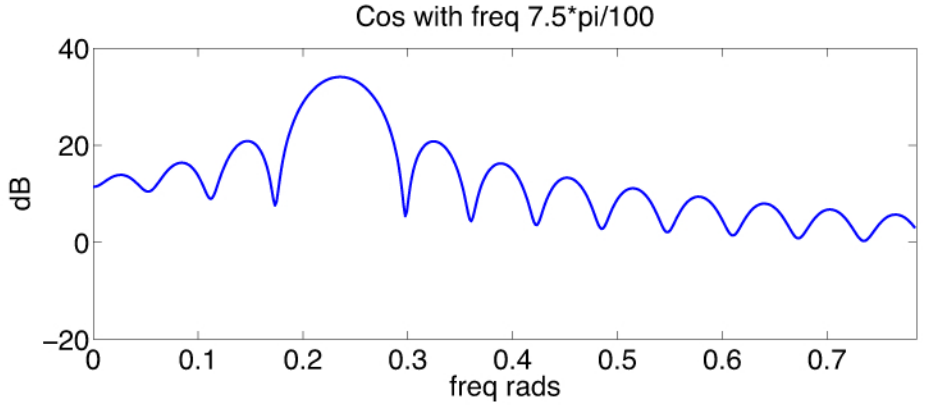
$|Z(e^{j\omega})|$

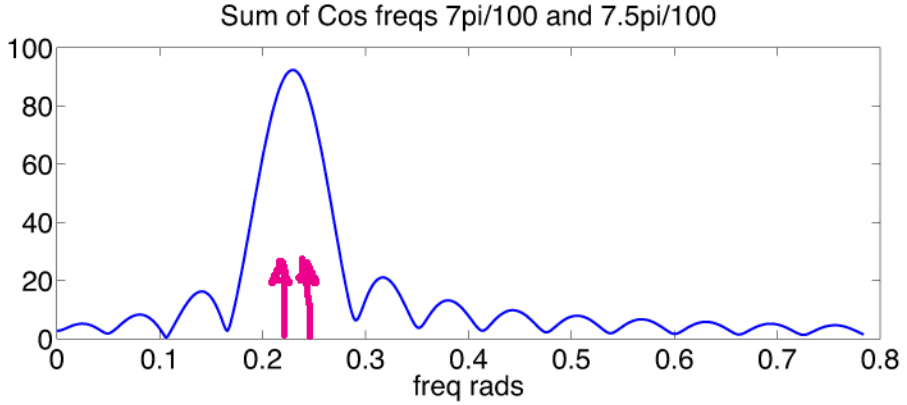
$N = 100$

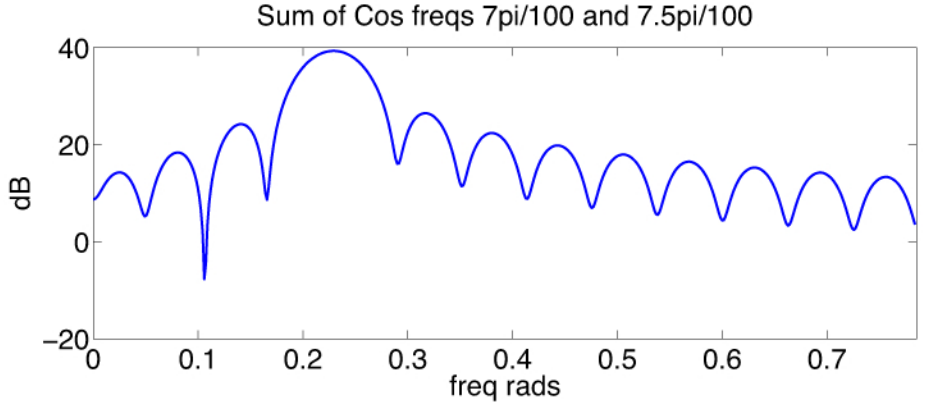








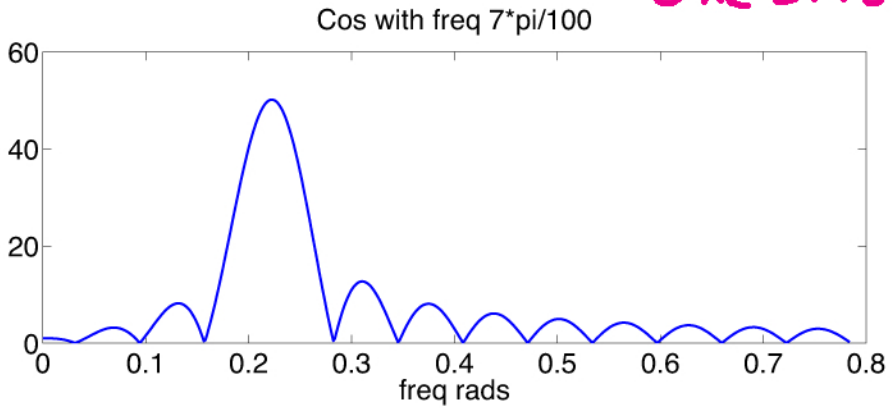


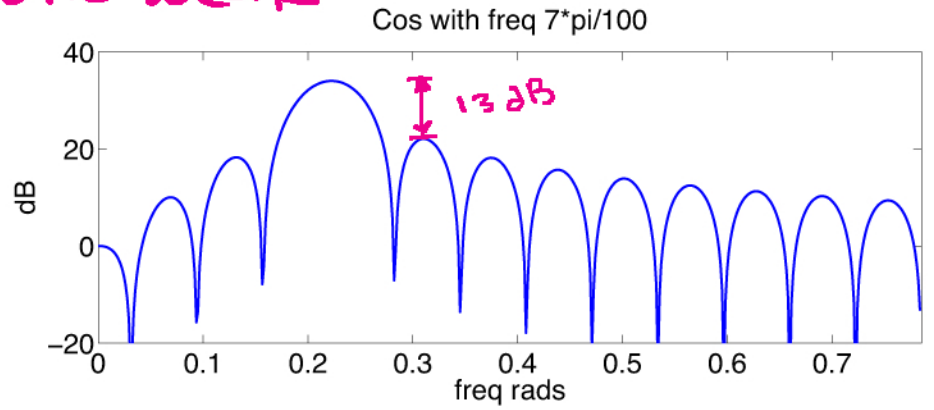


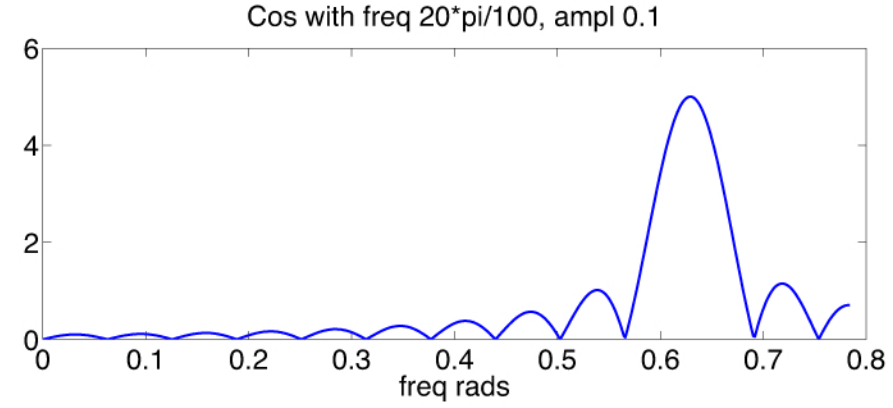
$|Z(e^{iw})|$

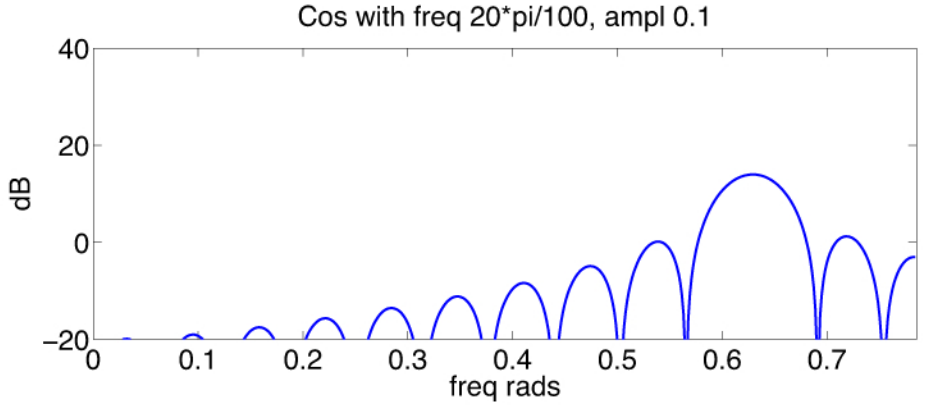
N=100

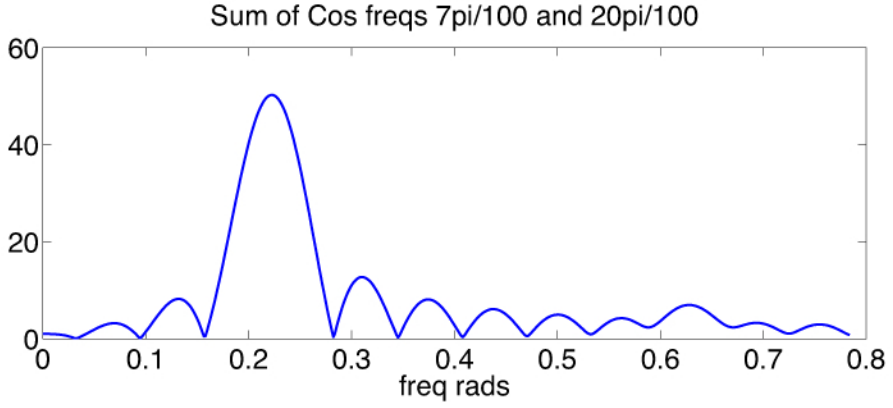
One strong, one weak

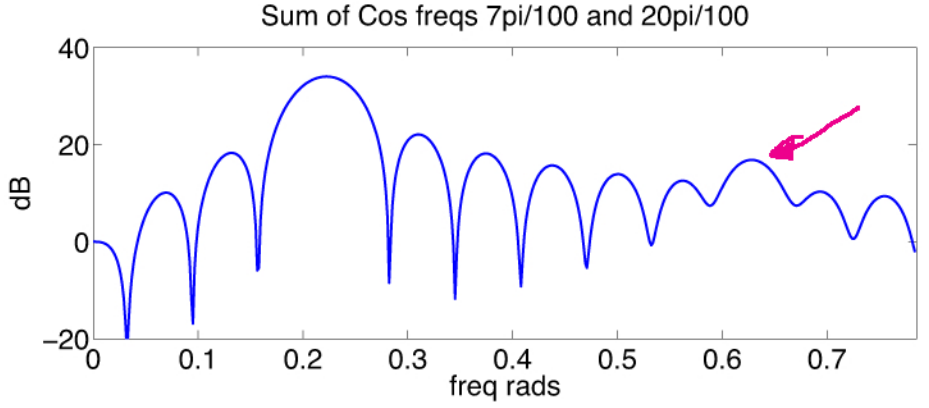








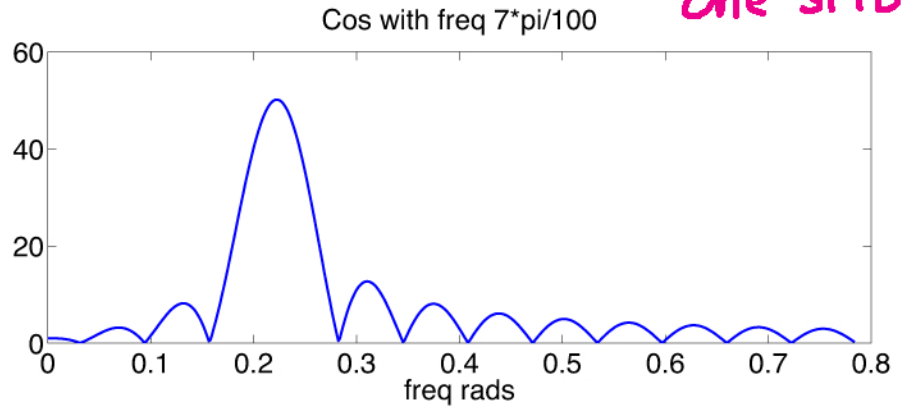


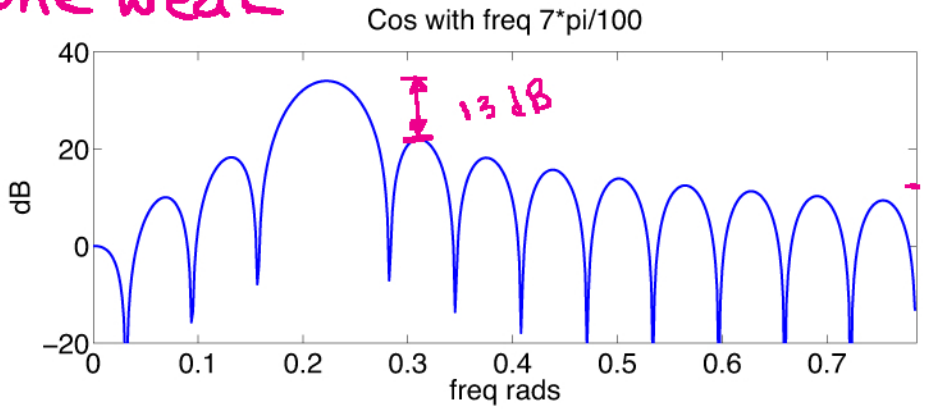


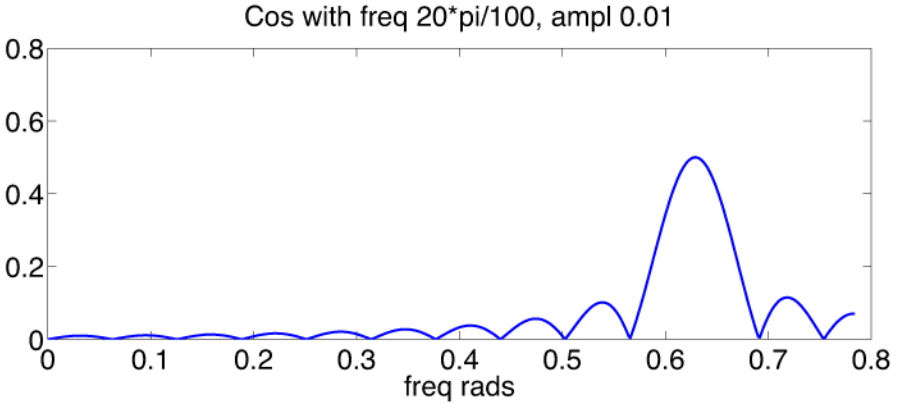
$|z(e^{j\omega})|$

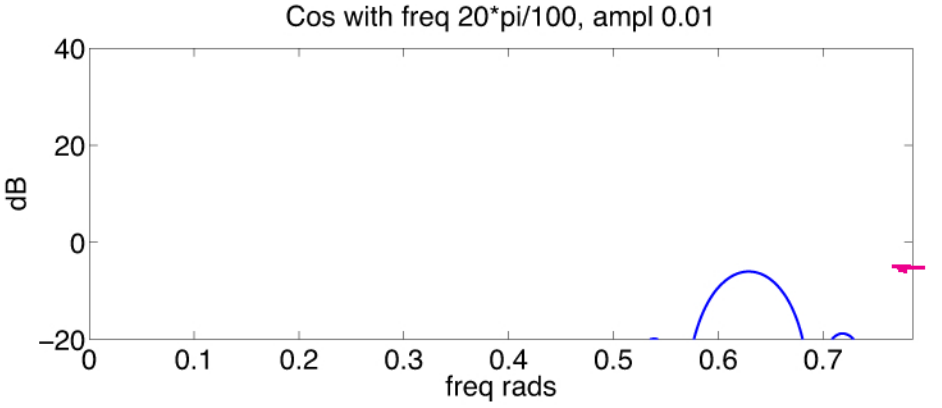
$N=100$

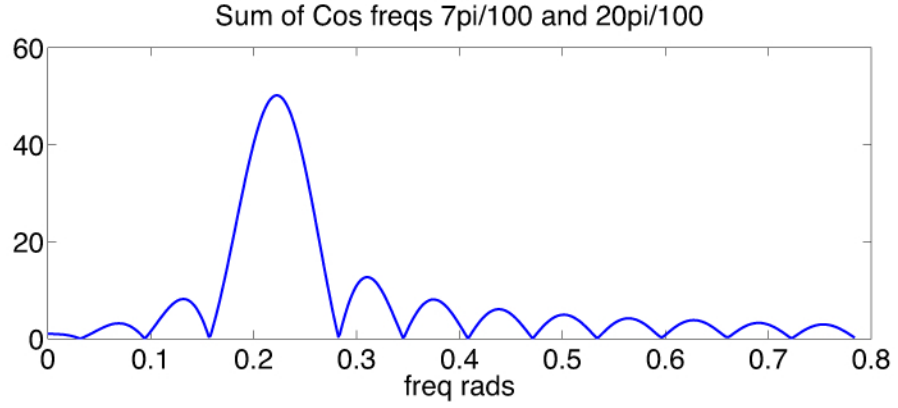
one strong, one weak

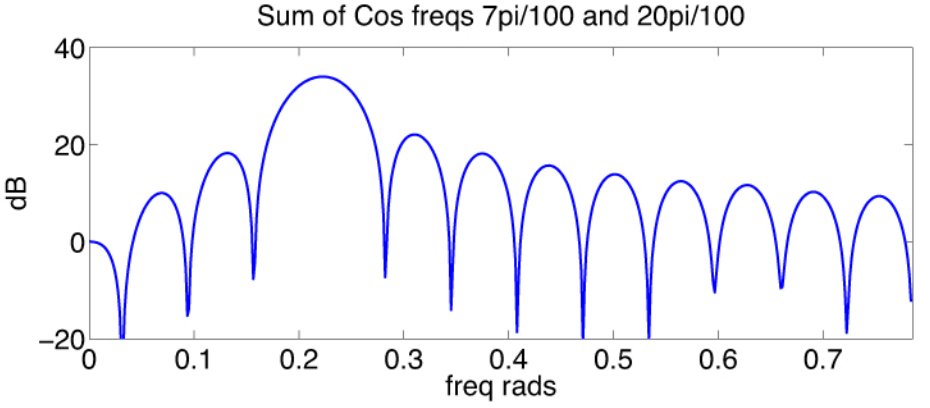












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